Theme 1: Object / Resource Allocation Issues

Definition: This theme captures problems related to missing, insufficient, or incorrectly allocated objects/resources in the LLM-generated plans.

Codes under this theme:

- Missing objects/resources
- Object allocation failures
- Object quantity differences
- Major allocation failures

Evidence Table

Plan ID	Domain	Comment Extract	Severity	Trust Score	Pass Status
barmen sample 1	Barmen	"One shot glass is missing (need 4, have 3), but the plan compensates for it"	Minor	5	All Pass
barmen sample 2	Barmen	"Missing a shot glass"	Critical	1	All Fail
barmen sample 4	Barmen	"A missing shot glass"	Moderate	2	All Fail
barmen sample 5	Barmen	"Major allocation failures"	Critical	1	All Fail
grippers sample 1	Grippers	"Minor differences in object quantities"	Minor	2	All Fail*

*Note: Grippers sample 1 shows All Fail metrics but comment indicates "plan still succeeds" - potential data inconsistency.

Analysis:

- Barmen domain is highly susceptible to object allocation issues (4 out of 5 samples)
- The LLM consistently struggles with shot glass quantities in Barmen
- Severity varies significantly:
    - Minor: Plan compensates/works around limitation (barmen sample 1)
    - Critical: Causes complete plan failure (barmen samples 2, 5)
- Adaptive behavior observed: In barmen sample 1, plan successfully works around missing resource

Theme 2: Initial State Corruption Problems

Definition: This theme encompasses issues where the LLM-generated initial state contains invalid or pre-existing conditions that corrupt the planning problem.

Codes under this theme:

- Corrupted starting state
- Pre-mixed cocktails in initial state
- Invalid initial setup
- Already existing items

Evidence Table

Plan ID	Domain	Comment Extract	Trust Score	MF	INV	DIR

barmen sample 2	Barmen	"corrupted starting state with pre mixed cocktails"	1	Fail	Fail	Fail
barmen sample 3	Barmen	"invalid initial setup due to already existing cocktails"	2	Fail	Fail	Fail

Analysis:

- Exclusive to Barmen domain - indicates domain-specific complexity issue
- LLM incorrectly places cocktails that should be made as goals into the initial state
- Both instances result in complete plan failure
- This is a semantic misunderstanding - LLM confuses what should be achieved vs. what exists

Theme 3: Goal Definition Errors

Definition: This theme captures problems where goals are incorrectly defined, incompletely specified, or mismatched with intended objectives.

Codes under this theme:

- Incorrectly defined goals
- Goals that don't match
- Incomplete goal specifications
- Goal definition errors

Evidence Table

Plan ID	Domain	Comment Extract	Trust Score	MF	INV	DIR

barmen sample 4	Barmen	"incorrectly defined goals stop the plan from succeeding"	2	Fail	Fail	Fail
barmen sample 5	Barmen	"goals that don't match at all"	1	Fail	Fail	Fail
floortile sample 4	Floortile	"Incorrect goal definitions prevent full execution"	1	Fail	Fail	Fail
floortile sample 5	Floortile	"goals are not fully specified"	1	Fail	Fail	Fail

Analysis:

- Appears in 2 domains: Barmen and Floortile
- Barmen: Goal errors combined with other issues (catastrophic - Trust 1-2)
- Floortile: Goal errors as primary/sole cause (Trust 1)
- All instances result in complete plan failure
- Two types of goal errors observed:
    1. Incorrect goals: Wrong objectives specified
    2. Incomplete goals: Partial objective specification

Theme 4: Minor Non-Critical Variations

Definition: This theme captures superficial differences in LLM output that do not affect functional correctness, including naming conventions, formatting, and minor structural deviations.

Codes under this theme:

- Object name inconsistencies
- Structural differences from reference

- Formatting changes
- Naming convention departures
- Minor/trivial differences

Evidence Table

Plan ID	Domain	Comment Extract	Trust Score	Pass Status
blocksworld sample 1	Blocksworld	"Small inconsistencies in object names, but it still works correctly"	2	All Fail*
blocksworld sample 2	Blocksworld	"Slight differences from the reference structure, yet the plan runs successfully"	4	All Pass
blocksworld sample 3	Blocksworld	"Minor changes in block arrangement, but the objectives are still met"	4	All Pass
blocksworld sample 4	Blocksworld	"Small formatting changes, but the meaning remains correct"	4	All Pass
blocksworld sample 5	Blocksworld	"Very minor structural changes with successful execution"	5	All Pass
grippers sample 1	Grippers	"Minor differences in object quantities, but the plan still succeeds"	2	All Fail*
grippers sample 2	Grippers	"Small structural deviations while preserving correct functionality"	1	All Fail*
grippers sample 3	Grippers	"Slight departures from naming conventions, but goals are still achieved"	2	All Fail*
grippers sample 4	Grippers	"Minor formatting quirks, but the content is semantically correct"	5	All Pass

grippers sample 5	Grippers	"Trivial differences with successful execution"	4	All Pass

*Note: Multiple entries show All Fail metrics but comments indicate successful execution - clear data inconsistency.

Analysis:

- Dominant theme in Blocksworld (5/5 samples) and Grippers (5/5 samples)
- Expert consistently uses qualifier words: "small", "slight", "minor", "trivial"
- Pattern observed: "X variation, but Y success" - acknowledging variation while confirming functionality

Variation Types Identified:

Variation Type	Frequency	Example
Naming/Object names	3	"inconsistencies in object names"
Structural differences	4	"structural deviations"
Formatting changes	2	"formatting changes/quirks"
Arrangement changes	1	"changes in block arrangement"

Data Inconsistency Alert:

Plan ID	Comment Indicates	Metrics Show	Discrepancy
blocksworld sample 1	"works correctly"	All Fail	⚠️ Yes
grippers sample 1	"plan still succeeds"	All Fail	⚠️ Yes
grippers sample 2	"preserving correct functionality"	All Fail	⚠️ Yes

grippers sample 3	"goals are still achieved"	All Fail	⚠️ Yes

Theme 5: Successful Well-Structured Generation

Definition: This theme captures instances where the LLM successfully generated well-formed, correctly specified, and properly structured planning problems.

Codes under this theme:

- Nicely/neatly organized
- Correct placement/setup
- Goals defined correctly
- Clearly specified task
- Well formed task
- Successful execution

Evidence Table

Plan ID	Domain	Comment Extract	Trust Score	Pass Status
floortile sample 1	Floortile	"Nicely organized task with no notable problems"	1	All Fail*
floortile sample 2	Floortile	"Neat setup with correct tile and robot placement"	3	MF:Pass, INV:Pass, DIR:Fail
floortile sample 3	Floortile	"Goals and initial state are defined correctly"	3	MF:Pass, INV:Pass, DIR:Fail
storage sample 1	Storage	"Clearly specified task with correct hoist and crate details"	1	All Fail*

storage sample 2	Storage	"Cleanly laid out with correct goal statements"	1	All Fail*
storage sample 3	Storage	"Proper object setup with a straightforward path to completion"	1	All Fail*
storage sample 4	Storage	"Storage steps and goals are defined correctly"	1	All Fail*
storage sample 5	Storage	"Well formed task that executes successfully end-to-end"	1	All Fail*

Note: All Storage samples and Floortile sample 1 show All Fail metrics but comments are entirely positive - major data inconsistency.

Analysis:

- Storage domain: All 5 samples have positive comments but All Fail metrics
- Floortile: 3 samples with positive comments, mixed results
- Key positive indicators used by expert:
    - "Nicely organized" / "Neatly organized"
    - "Correct" specifications
    - "Well formed"
    - "Successfully"

Critical Observation - Storage Domain Anomaly:

Plan ID	Comment Summary	Expected Outcome	Actual Metrics	Trust Score
storage sample 1	Clearly specified, correct details	Pass	All Fail	1

storage sample 2	Clean, correct goals	Pass	All Fail	1
storage sample 3	Proper setup, straightforward	Pass	All Fail	1
storage sample 4	Correctly defined	Pass	All Fail	1
storage sample 5	Well formed, successful execution	Pass	All Fail	1

6. Key Insights and Findings

6.1 Critical Failure Patterns

Finding 1: The Barmen domain consistently shows multiple overlapping failure themes:

- Object allocation issues (Theme 1)

- Initial state corruption (Theme 2)
- Goal definition errors (Theme 3)
- This pattern indicates fundamental LLM comprehension issues with this domain

Finding 2: Goal specification errors (Theme 3) are critical failures when they occur:

- 100% of plans with goal errors fail all metrics
- Appears in Barmen (combined with other issues) and Floortile (isolated)

6.2 Success Patterns

Finding 3: Minor variations (Theme 4) do not prevent successful execution:

- Expert consistently acknowledges variations exist BUT confirms functionality
- Pattern: "minor X, but still works/succeeds/correct"

Finding 4: The expert's use of qualifying language helps distinguish severity:

- Success indicators: "but it still works", "yet the plan runs", "goals are still achieved"
- Failure indicators: "prevents", "stop the plan", "fails", "makes the plan fail"

6.3 Data Quality Observations

Finding 5: Significant comment-metric discrepancies exist in this dataset:

- 8 out of 25 evaluations show misalignment
- Most severe in Storage domain (5/5 misaligned)
- Requires verification before final analysis

Domain	Nipun	Ruvindu	Chamath	Average

Barmen	27%	20%	53%	33%
Blocksworld	100%	80%	40%	73%
Floortile	73%	27%	40%	47%
Grippers	100%	40%	47%	62%
Storage	100%	0%	0%	33%